

MERN Stack Internship

DAY 8 ASSIGNMENT REPORT

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Position : MERNStackINTERN

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MERN Stack Internship – Day 8 Task Report

Task Title

Task Kanban Tracker Component

Objective

The objective of this task was to develop a client-side reactive Kanban Board using React. The application manages tasks through three different stages: Todo, In Progress, and Completed. The project also uses browser localStorage to ensure that task data persists even after page refreshes.

Technologies Used

- ☐ React.js
- ☐ JavaScript (ES6)
- ☐ HTML5
- ☐ CSS3
- ☐ React Hooks (useState, useEffect)
- ☐ Browser localStorage

Features Implemented

1. Three Kanban Columns

The application contains three task management columns:

- ☐ Todo
- ☐ In Progress
- ☐ Completed

Each task is displayed as a card within its respective column.

2. Task Status Management

Users can move tasks between columns using action buttons.

Available actions include:

- ☐ Move Todo → In Progress
- ☐ Move In Progress → Completed
- ☐ Move In Progress → Todo
- ☐ Move Completed → In Progress

3. React State Management

React's useState hook was used to store and manage task data dynamically.

4. Data Persistence

The useEffect hook and localStorage were implemented to save all task updates automatically.

This ensures:

- ☐ Tasks remain available after browser refresh.
- ☐ User changes are preserved.

5. Responsive User Interface

A modern responsive layout was designed using CSS Flexbox with custom styling, hover effects, shadows, and color-coded columns.

Project Structure

src/

├── App.jsx

├── App.css

└── main.jsx

Implementation Steps

Step 1

Created a React project using Vite.

Step 2

Initialized task data using useState.

Step 3

Created three Kanban columns:

- ☐ Todo
- ☐ In Progress
- ☐ Completed

Step 4

Displayed tasks based on their current status using the `filter()` method.

Step 5

Implemented task movement functionality using a custom `moveTask()` function.

Step 6

Added conditional action buttons for moving tasks between columns.

Step 7

Integrated `localStorage` using `useEffect()` to store task data.

Step 8

Applied custom CSS styling to improve the user interface.

Code Highlights

React Hooks Used

- ☐ `useState()`
- ☐ `useEffect()`

Browser Storage

- ☐ `localStorage.setItem()`
- ☐ `localStorage.getItem()`

Array Methods

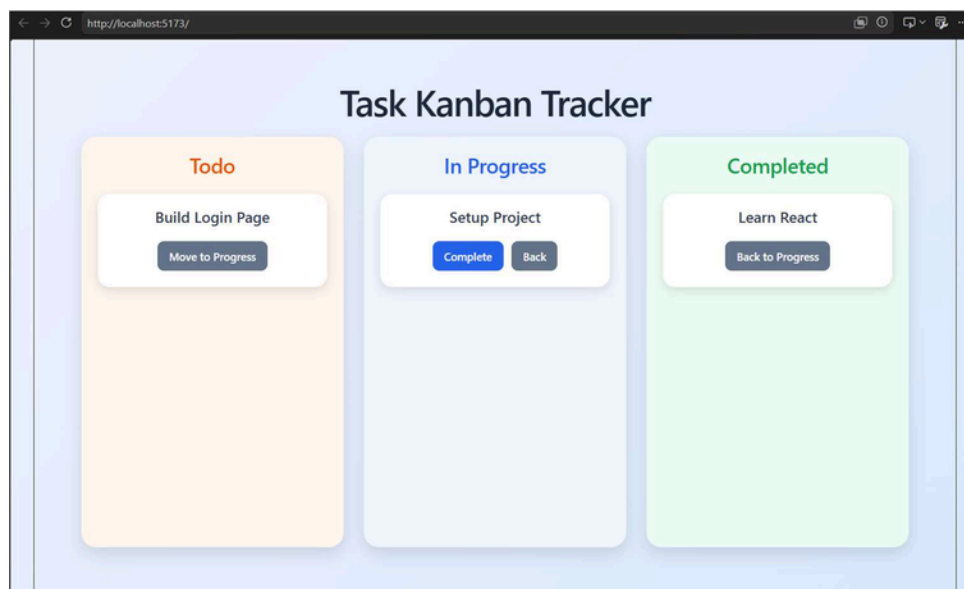
- ☐ `filter()`
- ☐ `map()`

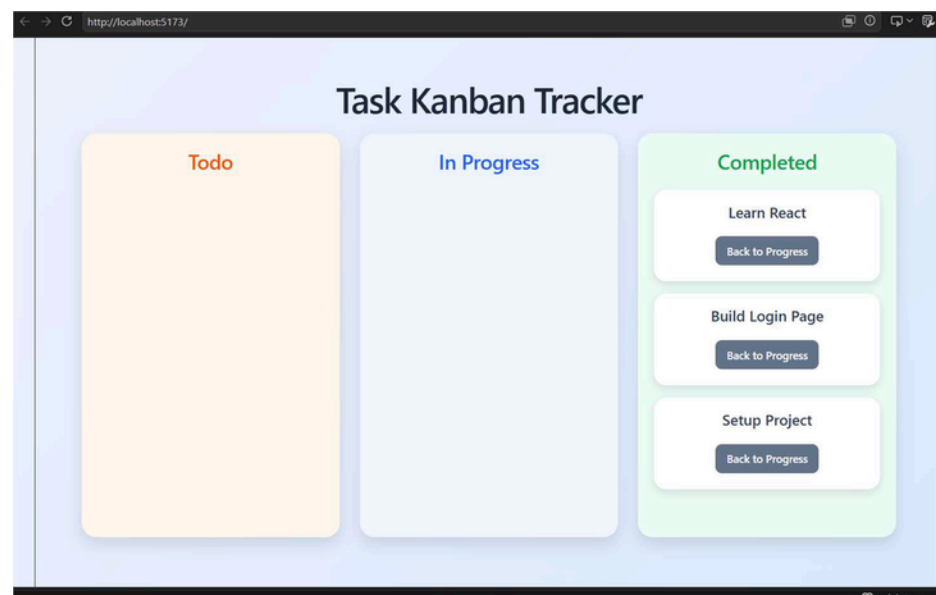
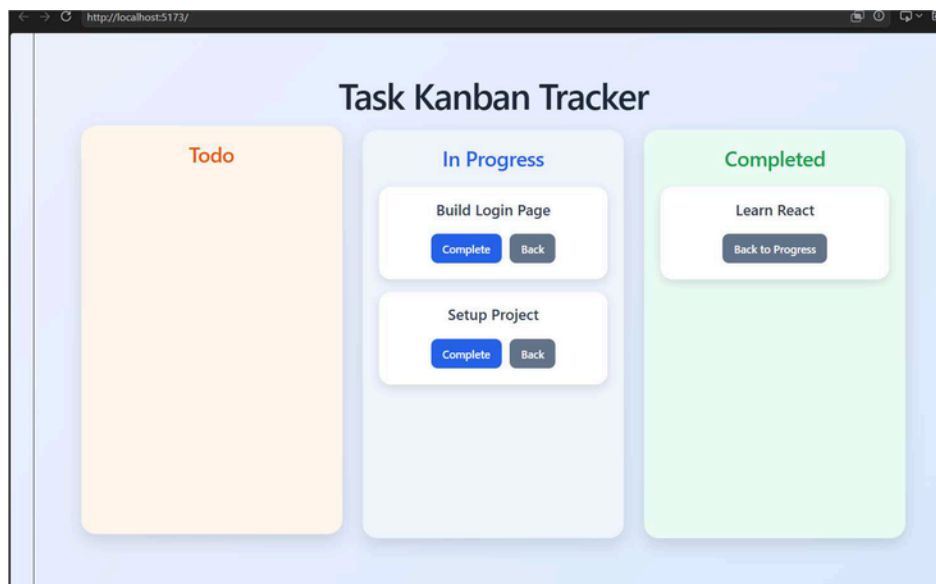
Learning Outcomes

Through this task I learned:

- ☐ React Functional Components
 - ☐ React State Management using useState
 - ☐ Side Effects using useEffect
 - ☐ Browser localStorage
 - ☐ Conditional Rendering
 - ☐ Dynamic UI Updates
 - ☐ Component-Based Development
 - ☐ Responsive CSS Design
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Output Screenshots





Conclusion

The Task Kanban Tracker was successfully developed using React and modern functional hooks. The application allows users to manage tasks through different workflow stages while maintaining task data using `localStorage`. The project demonstrates practical implementation of React state management, conditional rendering, and browser-based persistence mechanisms.